

Demonstration of Proposed Features

Date	13 July 2026
Team ID	AIML-HDI-2026
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Demonstration of Proposed Features

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S. No	Feature Name	Description	Status	Demo'd	Remarks
1	Country Selection & Auto-fill	Dropdown of 141 countries that auto-fills historical/rank data via the /country_data endpoint.	Implemented	Yes	Solves the issue of users not having historical HDI values on hand.
2	What-if Input Form	Editable Life Expectancy, Schooling, and GNI fields for exploring hypothetical scenarios.	Implemented	Yes	Fully functional on the live Flask form.
3	Input Validation	Server-side handling of invalid or missing numeric input before processing.	Implemented	Yes	Returns a friendly message instead of crashing the backend.
4	Data Normalization	Min-Max scaling of raw inputs to match the model's training-time distribution.	Implemented	Yes	Runs invisibly using the embedded scaling parameters.
5	Hybrid Neural Network Prediction Engine	Executes the dual-branch EnhancedHybridHDINet (PyTorch) to compute the HDI score.	Implemented	Yes	Verified against real countries (e.g., Norway: predicted 0.957 vs.

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					actual 0.961).
6	Results Display with Tier Badge	Displays the predicted HDI score and Low/Medium/High/Very High tier.	Implemented	Yes	Renders instantly via the /predict route.
7	REST Endpoints (/predict, /country_data)	Flask routes that receive form/JSON data and return predictions or lookup data.	Implemented	Yes	Verified via direct curl testing during development .
8	Persistent Public Deployment	Hosting the app so it stays live independent of the Kaggle notebook session.	Implemented	Yes	Deployed on Render at hdi-webapp.onrender.com — stays live independent of Kaggle or any local machine.

Feature Implementation Summary

Metric	Count / Value
Total Features Proposed	8
Total Features Implemented	8
Total Features Demonstrated	8
Overall Implementation Rate (%)	100%